

VIEW

Simple view with single table:

```
mysql> use employee;
Database changed
mysql> create table employee (emp_id int, emp_name varchar(30), salary int, department varchar(20));
ERROR 1050 (42S01): Table 'employee' already exists
mysql> drop table employee;
Query OK, 0 rows affected (0.07 sec)

mysql> create table employee (emp_id int, emp_name varchar(30), salary int, department varchar(20));
Query OK, 0 rows affected (0.04 sec)

mysql> insert into employee values(101,"Ravi",50000,"HR");
Query OK, 1 row affected (0.01 sec)

mysql> insert into employee values(102,"priya",60000,"IT");
Query OK, 1 row affected (0.01 sec)

mysql> insert into employee values(103,'Arun',45000,'sales');
Query OK, 1 row affected (0.00 sec)

mysql> insert into employee values(104,'Raju',70000,'Finance');
Query OK, 1 row affected (0.01 sec)

mysql> select * from employee;
+-----+-----+-----+-----+
| emp_id | emp_name | salary | department |
+-----+-----+-----+-----+
| 101 | Ravi | 50000 | HR |
| 102 | priya | 60000 | IT |
| 103 | Arun | 45000 | sales |
| 104 | Raju | 70000 | Finance |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> commit;
Query OK, 0 rows affected (0.00 sec)

mysql> create view v1 as select emp_id,salary where employee;
ERROR 1054 (42S22): Unknown column 'emp_id' in 'field list'
mysql> create view v1 as select emp_id,salary from employee;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from v1;
+-----+-----+
| emp_id | salary |
+-----+-----+
| 101 | 50000 |
| 102 | 60000 |
| 103 | 45000 |
| 104 | 70000 |
+-----+-----+
4 rows in set (0.01 sec)
```

Simple view with two tables:

```
mysql> select * from employee;
+-----+-----+-----+-----+-----+
| emp_id | emp_name | salary | department | dept_id |
+-----+-----+-----+-----+-----+
| 101 | Ravi | 50000 | HR | 10 |
| 102 | priya | 60000 | IT | 20 |
| 103 | Arun | 45000 | sales | 30 |
| 104 | Raju | 70000 | Finance | 40 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> create table department(dept_id int,department varchar(20));
ERROR 1050 (42S01): Table 'department' already exists
mysql> create table department1(dept_id int,department varchar(20));
Query OK, 0 rows affected (0.02 sec)

mysql> insert into department1 values(10,"HR");
Query OK, 1 row affected (0.01 sec)

mysql> insert into department1 values(20,'IT');
Query OK, 1 row affected (0.01 sec)

mysql> insert into department1 values(30,'sales');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into department1 values(40,'Finance');
Query OK, 1 row affected (0.01 sec)

mysql> insert into department1 values(40,'Finance');
Query OK, 1 row affected (0.01 sec)

mysql> select * from department1;
+-----+-----+
| dept_id | department |
+-----+-----+
| 10 | HR |
| 20 | IT |
| 30 | sales |
| 40 | Finance |
+-----+-----+
4 rows in set (0.00 sec)

mysql> select * from employee;
+-----+-----+-----+-----+-----+
| emp_id | emp_name | salary | department | dept_id |
+-----+-----+-----+-----+-----+
| 101 | Ravi | 50000 | HR | 10 |
| 102 | priya | 60000 | IT | 20 |
| 103 | Arun | 45000 | sales | 30 |
| 104 | Raju | 70000 | Finance | 40 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> create view v2 as select
-> e.emp_id,
-> e.emp_name,
-> e.salary,
```

```
-> e.emp_name,
-> e.salary
-> from employee e
-> join department1 d on e.dept_id = d.dept_id;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from v2;
+-----+-----+-----+
| emp_id | emp_name | salary |
+-----+-----+-----+
| 101 | Ravi | 50000 |
| 102 | priya | 60000 |
| 103 | Arun | 45000 |
| 104 | Raju | 70000 |
+-----+-----+-----+
4 rows in set (0.01 sec)
```

Complex view with single table:

```
mysql> select * from employee;
+-----+-----+-----+-----+-----+
| emp_id | emp_name | salary | department | dept_id |
+-----+-----+-----+-----+-----+
| 101 | Ravi | 50000 | HR | 10 |
| 102 | priya | 60000 | IT | 20 |
| 103 | Arun | 45000 | sales | 30 |
| 104 | Raju | 70000 | Finance | 40 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> insert into employee values(105,'kiran',55000,'HR',50);
Query OK, 1 row affected (0.01 sec)

mysql> commit;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from employee;
+-----+-----+-----+-----+-----+
| emp_id | emp_name | salary | department | dept_id |
+-----+-----+-----+-----+-----+
| 101 | Ravi | 50000 | HR | 10 |
| 102 | priya | 60000 | IT | 20 |
| 103 | Arun | 45000 | sales | 30 |
| 104 | Raju | 70000 | Finance | 40 |
| 105 | kiran | 55000 | HR | 50 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> create view v3 as select department , avg
-> (salary) as avg_salary from employee group by department;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from v3;
+-----+-----+
| department | avg_salary |
+-----+-----+
| HR | 52500.0000 |
| IT | 60000.0000 |
| sales | 45000.0000 |
| Finance | 70000.0000 |
+-----+-----+
4 rows in set (0.01 sec)
```

Complex view with two tables:

```
mysql> select * from employee;
+-----+-----+-----+-----+-----+
| emp_id | emp_name | salary | department | dept_id |
+-----+-----+-----+-----+-----+
| 101    | Ravi     | 50000  | HR         | 10      |
| 102    | priya    | 60000  | IT         | 20      |
| 103    | Arun     | 45000  | sales      | 30      |
| 104    | Raju     | 70000  | IT         | 20      |
| 105    | kiran    | 55000  | HR         | 10      |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select * from department1;
+-----+-----+
| dept_id | department |
+-----+-----+
| 10      | HR         |
| 20      | IT         |
| 30      | sales      |
| 40      | Finance    |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> select * from department1;
+-----+-----+
| dept_id | department |
+-----+-----+
| 10      | HR         |
| 20      | IT         |
| 30      | sales      |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> commit;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> create view v4 as select
-> d.department, avg(e.salary) as avg_salary
-> from employee e
-> join department1 d
-> on e.dept_id = d.dept_id
-> group by department;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from v4;
+-----+-----+
| department | avg_salary |
+-----+-----+
| HR         | 52500.0000 |
| IT         | 65000.0000 |
| sales      | 45000.0000 |
+-----+-----+
3 rows in set (0.00 sec)
```